Data Sheet

EXPERIMENT NO. 06

Title: - Simulation of Inverting and Non-Inverting Amplifier using Op-Amp (using Tinkercad/ Proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity

Important Formulae: -

Inverting Amplifier: -

$Vo / Vi = - Rf / R$

$Vo = - (Rf / R) \times Vi$

Non-Inverting Amplifier: -

$Vo / Vi = 1 + (Rf / R)$

$Vo = [1 + (Rf / R)] \times Vi$

- Vi = Input voltage
- Vo = Output voltage
- R = Input resistance
- Rf = Feedback resistance

Observation Table: -

Table 1: Inverting Amplifier — $Vo = - (Rf / R) \times Vi$

Sl. No.	Rf	R	Vi	Vo (Calculated)	Vo (Measured)

Table 2: Non-Inverting Amplifier — $Vo = [1 + (Rf / R)] \times Vi$

Sl. No.	Rf	R	Vi	Vo (Calculated)	Vo (Measured)

Calculation: -

Inverting Amplifier: Vo (Calculated) = − (Rf/R) × Vi = _____

Non-Inverting Amplifier: Vo (Calculated) = [1 + (Rf/R)] × Vi = _____

% Error between Calculated and Measured Vo = _____ %

Conclusion: -

Teacher's Sign and Date

Data Sheet

EXPERIMENT NO. 07

Title: - Simulation of Differentiator and Integrator Circuit using Op-Amp (using Tinkercad/ Proteus software circuit).

Instrument & Component Required for Differentiator: -

Sl. No.	Apparatus / Component	Specification	Quantity

Observation Table for Differentiator: -

I/P Waveform	P-P Amplitude of I/P (V)	I/P Time Period (ms)	P-P Amplitude of O/P (V)	O/P Time Period (ms)

Instrument & Component Required for Integrator: -

Sl. No.	Apparatus / Component	Specification	Quantity

Observation Table for Integrator: -

I/P Waveform	P-P Amplitude of I/P (V)	I/P Time Period (ms)	P-P Amplitude of O/P (V)	O/P Time Period (ms)

Important Formulae: -

Differentiator: -

$Vo = - Rf \times C1 \times (dVi / dt)$

Integrator: -

$Vo = - (1 / R1Cf) \times \int Vi\ dt$

- Rf, R1 = Resistance values
- C1, Cf = Capacitance values
- Vi = Input voltage, Vo = Output voltage

Calculation: -

Differentiator Output, $Vo = - Rf\ C1\ (dVi/dt)$ = _____

Integrator Output, $Vo = - (1/R1Cf) \int Vi\ dt$ = _____

Conclusion: -

Teacher's Sign and Date